

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****April-May 2015****EC310.01 Optical Communication****Date: 09.05.2015, Saturday****Time: 10:00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.****[07]**

- a. Light is confined within the core of a simple optical fiber by _____.
a) Refraction.
b) Total internal reflection at the outer edge of the cladding.
c) Total internal reflection at the core cladding boundary.
d) Reflection from the fiber's plastic coating.
- b. What makes fiber optic immune to EMI?
- c. State the difference between reflection and refraction.
- d. Snell's law relates _____.
a) Light reflection
b) Light refraction.
c) Light transmission.
d) Light absorption.
- e. The core of an optical fiber has a _____.
a) Lower refractive index than cladding.
b) Higher refractive index than cladding.
c) Similar refractive index with the cladding.
- f. What is material dispersion?
- g. Define bulk recombination lifetime.

Q – 2.a A step index fiber has a normalized frequency $V = 26.6$ at a 1300 nm wavelength. If a core radius is 25 μm , then find the numerical aperture. **[04]**

Q – 2.b Answer any two questions. [10]

- 1) Define Bending loss and also explain types of bending loss occur in single mode fiber.
- 2) What is Numerical Aperture? Derive equation for the same.
- 3) Discuss the most common type of mechanical misalignment occurring between two joined fiber and its effect.

Q - 3 Answer the following.

- a. A double- heterojunction InGaAsP LED emitting at a peak wavelength of 1310 nm has radiative and nonradiative recombination times of 30 and 100ns. find the quantum efficiency and internal power level for the drive current of 40mA. [05]
- b.
 - 1) Describe the construction of edge emitting LED with neat sketch. [05]
 - 2) Explain lensing scheme to provide coupling between the source and diode [04]

OR

- 1) Explain Fabry Perot resonator cavity for LASER diode. [05]
- 2) Explain different fiber splicing technique. [04]

SECTION – II

Q - 4 Answer the question below. [07]

- a. Define photo current .
- b. Define impact ionization.
- c. Which of the following is fastest light sensor?
- a) APD.
 - b) PIN.
 - c) Photo diode.
 - d) Photovoltaic diode.
- d. What is packet switching?
- e. DWDM stands for _____.
- f. Slicing fiber means _____.
 - a) Fusion.
 - b) Butt.
 - c) Glue.
 - d) Both fusion and butt.
- g. What is surface leakage current?

Q – 5.a A given Silicon avalanche photodiode has a quantum efficiency of 65% at a wavelength of 900 nm. Suppose 0.5 μW of optical power produce a multiplied photocurrent of 10 μA . Then find out the multiplication factor M . ($q = 1.6 \times 10^{-19} \text{ C}$, $h = 6.625 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ m/s}$) **[05]**

OR

Q – 5.a An engineer has the following component available: **[05]**

- a) GaAlAs laser diode capable of coupling 1mw at 850 nm.
- b) Ten section of cable each of which is 500 m long, as a 4-dB/km attenuation and has connector on both ends.
- c) Connector loss of 2db/connector.
- d) A *pin* photodiode and APD as a receiver

Using these components, the engineer wishes to construct a 5-km link operating at 20 Mb/s. If the sensitivity of the *pin* and APD receivers are -45 and -56 dBm respectively, which receiver should be used if a 6 –db system operating margin is required?

Q – 5.b Explain quantum efficiency and responsivity of photodiode. **[04]**

Q – 5.c Write a short note on photo detector noise. **[05]**

OR

Q – 5.c Explain principal of operation, the structure and the characteristic of APD. **[05]**

Q – 6.a Explain subcarrier multiplexing. **[04]**

Q – 6.b Explain amplification mechanism with reference to EDFA. **[07]**

Q – 6.c What are the key system features of WDM? **[03]**

OR

Q – 6.a Explain 2 x 2 fiber coupler. What are the different losses associated with it? **[06]**

Q – 6.b Draw and Explain SONET frame structure. **[06]**

Q – 6.c Draw schematic diagram of a typical optical receiver. **[02]**
